PSTN/ISDN Platform integration — a case history

S T Marsden

This paper describes the service platform systems integration approach adopted for number portability, and explains how this helped to ensure that the set of systems and operational processes which make up the PSTN/ISDN Platform work together seamlessly to deliver the number portability service. The paper then discusses how the service platform approach to systems integration is being used to expedite the successful delivery of other services on the PSTN/ISDN Platform.

1. Introduction — service platforms

The BT platform framework defines a service platform as:

> *'The integrated set of components, including network elements, operational support systems and operational processes which, taken together, provide an environment in which sellable BT services can be delivered.'*

Telephone exchanges and transmission systems are types of network element, and order handling and billing systems are typical examples of operational support systems (OSS). Operational processes are the sets of activities (or 'workstrings') that must be performed by BT people to provide world-class service to customers. Examples are provision, billing and repair processes.

A service platform is much more than a complex assembly of telecommunications systems. The inclusion of operational processes transforms this set of sophisticated, but essentially inanimate, technology into a living part of the business entity that is BT.

2. Service platform integration

Systems integration in BT is a process which ensures that a number of telecommunications and computer systems work together seamlessly to support a range of business functions. Figure 1 shows how service platform integration extends this process by ensuring that operational processes work properly with the OSS contained in the platform. The results of this work are very relevant to BT's Networks and Service Operations (N&SO) gatekeeping process.

The component systems depicted by the four boxes at the foot of Fig 1 are likely to be created from smaller systems which have been subject to verification, validation and testing (VV&T) throughout their development cycle. However, this will be limited in scope to the component systems themselves and cannot address end-to-end service platform issues. The service platform integration process ensures that early life cycle activity [1] such as requirement analysis and design verification focus on the ability of these systems to work together and with the relevant operational processes. The same is true of the testing performed when the component systems are delivered into the platform integration environment.

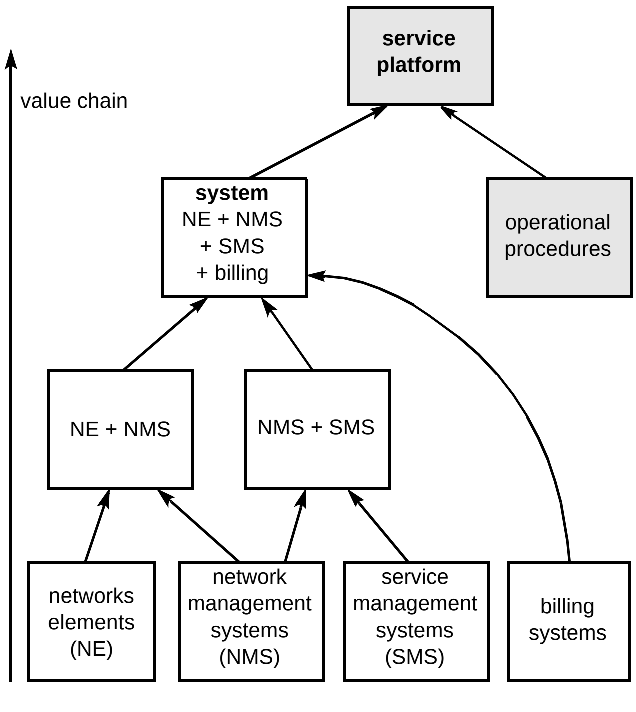

Fig 1 Service platform integration.

The next two boxes show how network elements (NE) might initially be integrated with network management systems (NMS) and NMS with service management systems (SMS). The VV&T performed during this process ensures that these component systems work together, complementing the in-depth VV&T performed by the component systems integration teams, which ensures that the component systems function properly in isolation.

The next step might be to integrate the billing systems with the NE, NMS and SMS to create the complete system. Again, early life cycle work is performed to ensure that when these systems are tested together for the first time, few faults are likely to emerge.

The shaded boxes in Fig 1 show how the operational processes are integrated with the complete system to form a service platform. Once the operational processes have themselves been verified (see section 7 for more details of this process), they can be tested end-to-end across the complete platform. Although generic parts of the operational processes may have been tested during component system integration, it is only at this stage that product-specific end-to-end testing can take place.

After end-to-end platform testing has been successfully completed, other activities such as operational readiness testing and technology trials may be required before the products supported by the service platform can be launched.

The arrow shown on the left of Fig 1 shows how business value increases as more systems are brought together, until a platform is finally created for the delivery of revenue generating services. This does not suggest that the business value of stand-alone systems is low (indeed, the service platform could not be realised without them), but rather that their full potential is only delivered when they are used in conjunction with other systems and with operational processes.

The benefits of the service platform integration approach are that it:

- ensures that early life cycle work (e.g. requirements analysis and design verification) encompasses all of the platform components, reducing time to market by enabling problems to be resolved before any software has been developed or modified,
- facilitates platform ‘husbandry’ along the value chain, to complement the normal integration process for component systems,
- extends systems integration (SI) and VV&T discipline throughout the entire platform integration process, to ensure that maximum business benefit is delivered within reduced time-scales, and at least cost,
- ensures that operational processes work properly,
- significantly reduces time-scales and cost by enabling VV&T and SI to be performed simultaneously for several products and services on a single version of a service platform.

3. Examples of service platform integration in BT

Large-scale service platform integration probably began in BT in 1990 with project Cyclone. This project integrated systems from both internal and external suppliers while ensuring that they supported the relevant operational processes, enabling BT to deliver virtual private network services on an intelligent network platform.

The national code change was another example of managing significant change across a large number of systems and operational processes. More recently, the creation of BT’s Card Services Platform required the integration of systems from various suppliers, and in this case it was necessary to ensure that operational processes of several external card suppliers were supported as well as BT’s own operational processes.

Service platform integration has also been performed within several interactive, on-line and multimedia projects. As a result of this approach, all of these projects have successfully delivered robust solutions on time. The next five sections of this paper explore one particularly large-scale PSTN/ISDN Platform integration project — Number Portability.

4. Number portability — the challenge

By the early 1990s, concern was growing among BT’s UK competitors that the need for customers to change their telephone number when moving from BT to an ‘other licensed operator’ (OLO) was a barrier to competition. Number portability removes this restriction and BT began discussing the requirements for this service with the UK telecommunications regulator, Oftel, and a number of UK public network operators within the Public Network Operator’s Interest Group (PNOIG). Unfortunately, attention initially focused on the network aspects and the PNOIG did not really address the operational support issues. BT’s operating licence allows six months for the deployment of network services specified by an Oftel determination, from the date that the necessary capabilities are delivered to BT by our switch suppliers. In such a large, distributed organisation, modifying existing OSS, and assessing whether any new OSS were necessary for number portability, posed a significant challenge.

BT’s initial demonstration of number portabilty took place in late 1994 using call diversion features on System X exchanges, together with a number of manual procedures. A more durable network solution was then envisaged requiring relatively small changes by the switch manufacturers (Ericsson and GPT). However, far more development was required for BT's OSS and operational processes.

BT has several hundred OSS and any of these that relied upon directory numbers might have needed changes for number portability. For example, BT's directory enquiry systems rely upon the first few digits of the full directory number to decide whether a customer belonged to BT or to an OLO. If the number belongs to an OLO, then the customer details (e.g. name and address) must not be changed by BT, and vice versa.

In particular there was a major impact on the customer service systems (CSS), which are widely used by BT's customer facing divisions and the operations and maintenance centre (OMC), which manages BT's exchanges.

BT faced a major task in having to assess the potential impact of number portability on all of its OSS. The approach adopted was to work through the key operational processes for number portability service delivery so that the impact upon all systems that would need to support these processes could be assessed. These processes were:

- order handling and billing,
- porting of numbers,
- fault reporting,
- service establishment.

The end-to-end and solution designs were essential aspects of this assessment. The Number Portability integration team helped the design team address this daunting change programme by devising a V&V technique which could ensure that the resultant processes and design met the requirements of the various business units across BT. The approach adopted was to hold role-playing process walk-throughs, and this will be described in more detail in section 6.2.

5. Number portability — solution overview

In number portability jargon, a customer is exported from one operator and imported by another.

The current solution for number portability involves inserting a 6-digit prefix into the signalling information associated with calls to an exported number. The prefix is inserted at the digital local exchange (DLE) which originally hosted that number, as shown in Fig 2. Different techniques are used to achieve this in different exchange types, but the result is to route calls via a digital main switching unit (DMSU) to the appropriate point of interconnect with the importing operator's network.

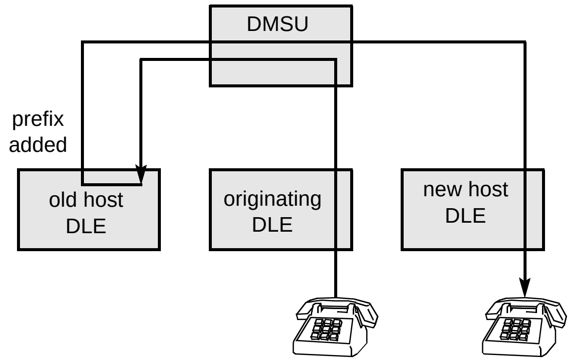

Fig 2 Number portability call 'tromboning'.

Calls to an exported number, from the network which originally hosted that number (but from a different DLE), are 'tromboned' as depicted in Fig 2. One problem associated with this solution is that the speech path remains routed through the DLE, thereby increasing network traffic. However, a 'call dropback' solution is in development that will minimise the impact on the network.

6. Number portability platform integration

The platform integration process developed for number portability involves the following stages (but work is not necessarily performed in this order):

- early life cycle work (verification and validation),
- operational process walk-throughs,
- ancillary system testing,
- end-to-end platform testing,
- operational technical testing (OTT),
- trial,
- pilot.

The expertise gained by the integration team has also enabled them to provide support for technical trials and service management centre and zone operations. Each of these stages will now be described in more detail.

6.1 Early life cycle work — verification and validation

Number portability involves enhancements to many existing systems. It is not a new system; neither does it require any major new systems to be developed. Number portability early life cycle work therefore involves the assessment of many system designs to verify the capability of the emergent platform to meet both industry and BT requirements for the number portability service [2].

The availability of component system documentation is essential for number portability early life cycle work and this is provided under agreements with component system design, development and VV&T teams. Number portability early life cycle work aims to achieve:

- monitoring of component system VV&T activity from an end-to-end viewpoint to ensure that the emergent platform complies with business requirements,
- continuous risk management during component system development and dynamic design activity,
- end-to-end operational process verification.

6.2 Operational process walk-throughs

Operational processes for number portability (order handling, fault and repair, billing and service establishment) are described in dynamic designs. Authority for these processes lies with BT's network and service operations and customer facing divisions. The processes are verified by means of structured process walk-throughs managed by the number portability integration manager and controlled by an accredited rapid application development (RAD) facilitator. Even when processes are thought to be stable, this approach frequently enables participants to identify serious problems that can then be remedied before product launch.

The walk-throughs revolve around predetermined test scenarios, which may also include provocative exception conditions. The normal flow of work is followed through the relevant process, with participants taking on previously agreed roles of customers, customer service agents, network and service operations people, etc.

It is important that people only work from existing processes or designs. For example, the people acting out the customer roles should only disclose the information they are asked for in the dialogue with the customer service agent. Where paper pro formas have to be passed between participants, these may be pre-filled with data relevant to the particular scenario to save time. As process issues, errors or changes arise, they are listed on whiteboards or flipcharts for all to see, and captured in detail on pro formas by the relevant participants.

Seating arrangements are quite important. People are deliberately arranged such that the work-flow moves back and forth across the table, so that all present can follow the activities being performed.

Participants are normally arranged as an inner and an outer circle. People in the inner circle have specific roles to perform and are knowledgeable users of the system or process, with the authority to make decisions. The people in the outer circle are either observers attending to learn, or experts who can be consulted when required. Observers are asked not to contribute directly, but are encouraged to log issues and comments on their own whiteboard. The walk-through then pauses when observers have logged a number of issues so that these can be discussed.

All participants have a chance to speak on equal terms since every part of the end-to-end process is given equal attention. Despite the fact that large meetings can be very difficult to manage with conventional techniques, feedback questionnaires show a high level of satisfaction for walk-throughs involving up to 50 people.

The sense of team play after the first few sessions is noticeable and people become very open, honest and motivated. These walk-throughs are also the first time many of the teams fully appreciate how adjacent teams will use the information passed on.

One particular measure of success is that participants are invariably keen to repeat the process until no errors remain — three full-day workshops were needed to get one number portabilty process right, but participants could see concrete results emerging and were only too keen to commit their time.

6.3 Ancillary system testing

Although number portability did not require development of new OSS, several PC-based ancillary systems were developed. In general, development employs RAD-style techniques and the VV&T approach followed is similar to that outlined in Carter et al [3]. These systems undergo integration and test in an independent environment prior to operational technical testing (OTT) and also form part of the end-to-end test environment.

6.4 Platform end-to-end testing

When component systems have been integrated and tested, they are integrated into the number portability platform test environment (NPTE) so that end-to-end platform testing can proceed. The NPTE design is fully documented in the NPTE User Guide and the 'Data Parity Definition for Number Portability End-to-End Testing'.

A schematic diagram of the NPTE is given in Fig 3. Neither the NPTE nor its component systems will be described in detail; this diagram is merely included to indicate the scope and scale of the NPTE.

The test plan contains three vital elements:

- a statement of intended test coverage,
- a checklist — ensures NPTE is correctly configured,
- the test schedule.

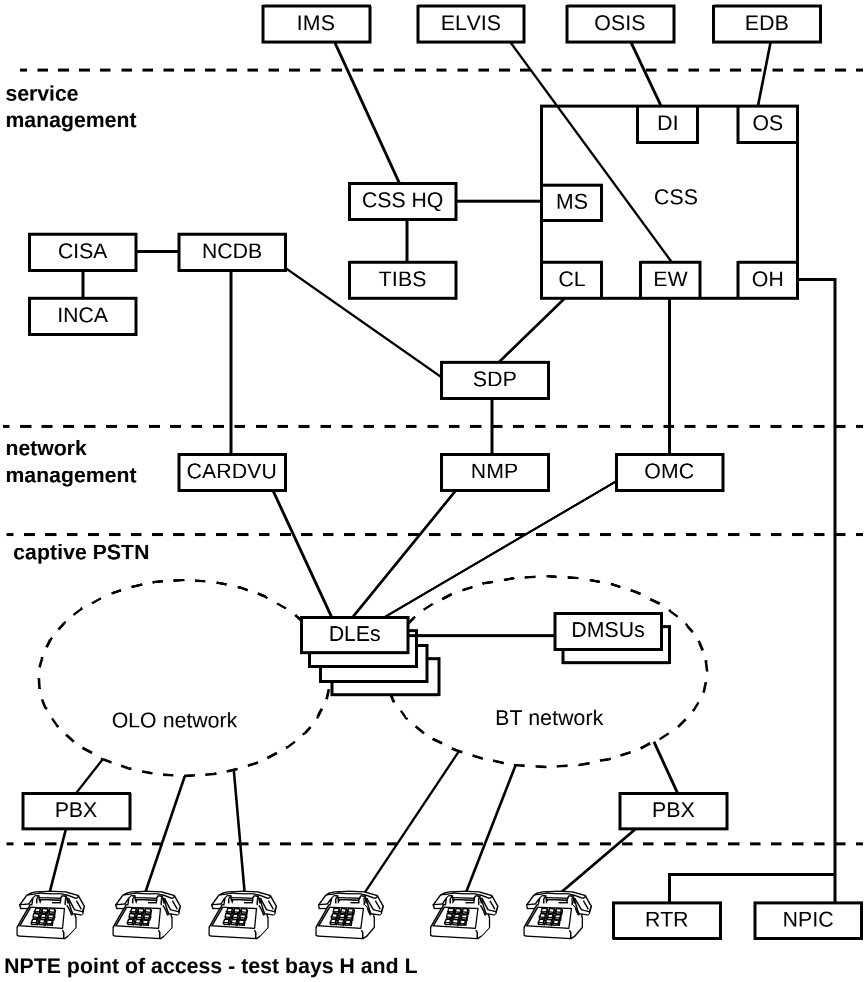

Fig 3 Number portability platform test environment.

In the early days of number portability VV&T, it could take up to six weeks to resolve the many problems which inevitably arise when modelling such a large and complex platform integration and test environment. However, use of the NPTE checklist as part of the end-to-end test quality gate has reduced this period to one or two weeks. Since the NPTE is too big to be owned by a single BT unit, the end-to-end test manager must be very proactive when liaising with other users to pull the environment together.

It is also important to ensure that the number portability end-to-end test schedule is sufficiently flexible that, if part of the NPTE fails, testing can still continue. This typically involves interrupting a test at a point where a system problem causes a break in a workstring, and storing the relevant test data. Other workstrings can then be tested before testing is resumed on the original workstring. To achieve this flexibility, however, the process for managing change in the platform test environment must be mature and fully effective.

Figure 4 is a simplified representation of the number portability platform end-to-end test process, which commences when all component systems have passed a stringent set of quality checks (referred to as a quality gate) to ensure that they are ready to enter end-to-end testing stage.

Number portability functionality is assessed across the platform, together with the relevant operational processes. The draft provision process is used, together with number portability requirements and the end-to-end design, as the basis for end-to-end test designs. Due to limitations of the test environment, fault and repair processes are not tested until operational readiness testing.

A test report is produced by the number portability integration manager on completion of end-to-end platform testing. This is a record of the extent to which the component systems provide the capabilities agreed in the scope of the release. Problems identified in both systems and processes are raised as problem reports as well as being listed in the test report, which therefore contains all of the information necessary for the quality gate prior to OTT.

Figure 4 shows how BT's major OSS (CSS and OMC) progress from their own system tests through pilot and into

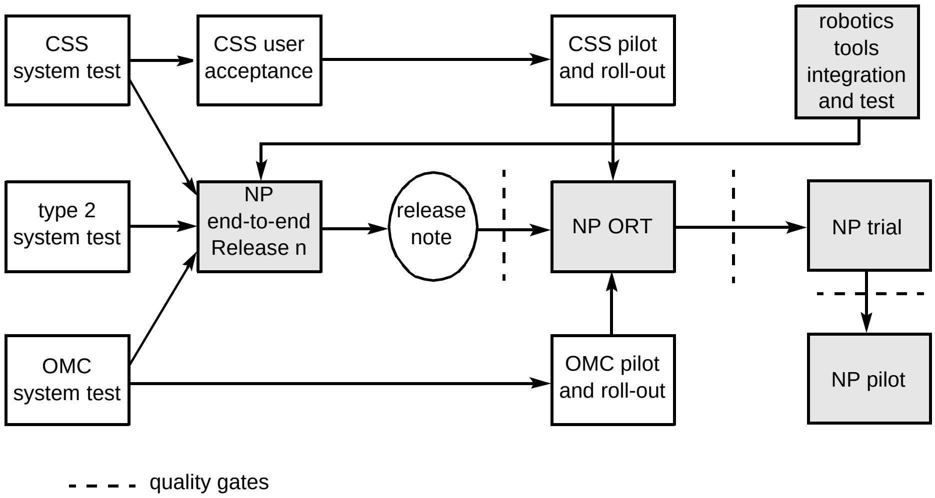

Fig 4 End-to-end testing process.

roll-out. Because of the many changes and improvements included in each release of these systems, no single product is on this critical path. These systems are included in the number portability test environment when the main system tests have been completed and the number portability, OMC and CSS integration teams agree that they are sufficiently stable.

Type 2 systems are those which do not need to change to enable the delivery of number portability, but which must interwork with systems that do. These are integrated together with OMC, CSS and ancillary systems to create the complete end-to-end test environment.

6.5 Operational technical testing

The purpose of OTT is to test the functionality and procedures in a real environment in collaboration with an OLO. The number portability service management centre is the central point of contact for number portability queries and plays a major role in the tests. OTT is co-ordinated by the platform integration team, with support from various other BT teams.

6.6 Number portability trial

The trial stage is intended to be a small-scale implementation of a number portability contract with an OLO in which a limited number of customers are ported.

6.7 Number portabilty pilot

The pilot stage is a larger scale implementation of number portability prior to full service establishment with the importing OLO.

7. Roles and responsibilities

The key roles that form the basis of the PSTN/ISDN Platform integration team (as described in the next section) are as follows:

- Integration Manager — responsible for all aspects of number portability integration and VV&T,
- VV&T Manager — with specific responsibility for VV&T throughout the life cycle,
- Platform Test Environment Manager — responsible for designing, implementing and maintaining the test environment,
- Platform Test Data Co-ordinator — ensures that test data is fully representative of operational data so that the test results are relevant in the operational environment,
- End-to-end Test Manager — manages the team which works with the Platform Test Environment Manager to ensure that the environment is ready, prior to performing end-to-end platform testing,
- Operational Technical Test Manager — with responsibility for co-ordinating OTT with the Number Portability Service Management Centre, the relevant OLO and other parts of BT involved.

8. PSTN/ISDN platform integration

This process builds upon the experience gained in many service platform integration projects, particularly number portability, as described in the previous sections.

8.1 Process

The PSTN/ISDN Platform integration process commences with an assessment of the system integration work required for the delivery of a PSTN/ISDN product, based upon the following considerations.

- Does the product require new operational procedures to be created?
- If existing procedures are to be used, do they need to be modified?
- Does the product depend upon process workstrings that traverse several OSS?
- Does the product require significant changes to OSS or network systems?
- Will new or modified data streams be required?

If the answer to any of these questions is yes, it is likely that a platform integration approach would be beneficial.

The next step is to assign an integration manager to the product team. This product integration manager (PIM) is normally an experienced systems integrator who will attend product team and Networks and Systems (N&S) delivery project meetings as required. To ensure continuity, the PIM will remain assigned to the product team until the product has been successfully launched, and will remain the primary point of contact for the relevant PSTN/ISDN product for systems integration issues.

PSTN/ISDN Platform integration work is based on a generic quality plan, a generic integration strategy and a generic VV&T policy. A single end-to-end test plan is produced for each release of the PSTN/ISDN Platform, together with product-specific test designs for sign-off by the relevant product team. A single end-to-end test campaign is conducted, covering all new product functionality for the relevant release of the PSTN/ISDN Platform, plus appropriate regression tests. A single test report is then produced with a generic section, plus product-specific sections for delivery to product teams.

Figure 5 shows how integration and end-to-end VV&T of the PSTN/ISDN Platform (including product-specific ancillary systems) can progress in parallel with work on the constituent major systems. Feedback paths for problem reports from PSTN/ISDN VV&T to the major systems delivery programmes have been omitted for clarity.

In Fig 5 OSS and network systems are incorporated into the PSTN/ISDN Platform integration environment when the bulk of their stand-alone testing has been successfully completed and they are sufficiently stable. To ensure that the PSTN/ISDN Platform integration team is able to judge when these component systems should be incorporated, close liaison is maintained with the component system test teams. Any ancillary systems are also integrated into the PSTN/ISDN Platform integration environment and end-to-end testing proceeds. Effective configuration management is required to ensure that any changes to the major component systems between their stand-alone system tests and roll-out are captured in the PSTN/ISDN Platform integration environment.

When end-to-end testing has been satisfactorily completed the platform progresses through any pre-trial tests (there may be separate tests for each product supported by the particular platform release) through trials and pilots

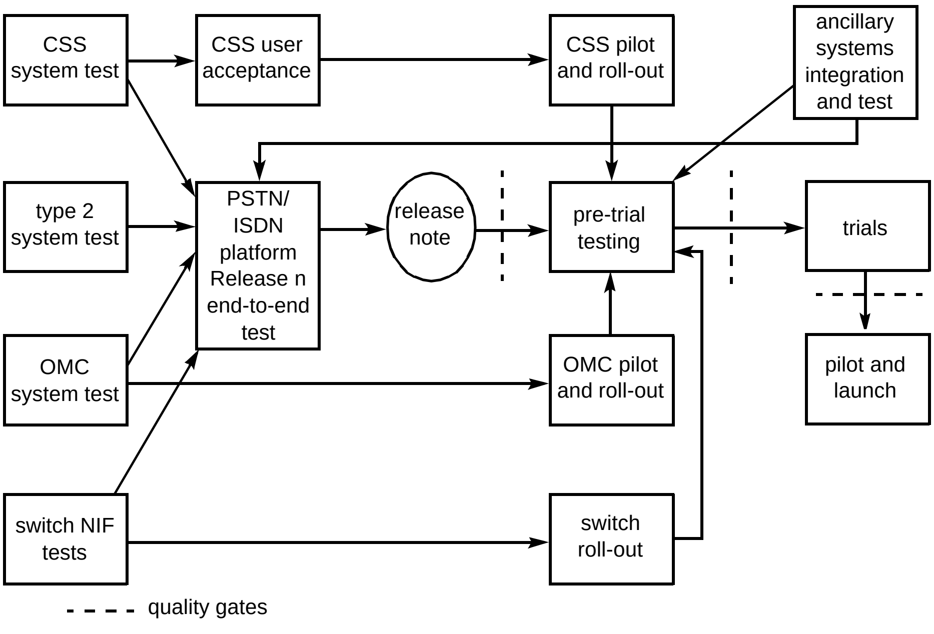

Fig 5 PSTN/ISDN integration (simplified).

to product launch. Throughout the whole process, N&SO gatekeepers and customer-facing division product launch managers are given full visibility as confidence in systems, processes, products and services increases.

It should be noted that the major component systems must be deployed before the products and services that depend upon these systems can be launched.

8.2 PSTN/ISDN platform integration team

This team is essentially the same as the number portability integration team, but with a few modifications to enable it to support several PSTN/ISDN products:

- PSTN/ISDN Platform Integration Manager (reports to PSTN/ISDN Programme Board),
- Product Integration Managers (one per product),
- VV&T Manager,
- Platform Test Environment Manager,
- End-to-end Test Manager,
- Platform Test Data Co-ordinator,
- several VV&T engineers to support Product Integration Managers as required.

The volume of work handled by this team is naturally regulated by the throughput of the CSS and OMC work-stacks. The in-depth VV&T performed by network and OSS integration teams enables the PSTN/ISDN integration team to focus more on breadth — ensuring systems and processes work together to support product launch and service delivery.

8.3 Current work

Several aspects of the number portability platform integration process are currently being applied to other PSTN/ISDN products and services. Work has also just begun on Year 2000 compliance for the PSTN/ISDN Platform. Early indications are that this process is now sufficiently mature to offer a generic approach to PSTN/ISDN Platform integration. An economy of scale has already been observed wherein test engineers with a significant amount of experience of number portability integration were able to perform end-to-end testing for another PSTN/ISDN product in two days, rather than the two weeks or so which would otherwise have been required. The more products integrated and tested onthe PSTN/ISDN Platform, the lower the cost per product and the shorter the duration of the work required for each product.

8.4 The future

As BT faces more competition in the PSTN/ISDN services market, and more demands from industry and the regulator, the platform integration approach will be widely adopted, otherwise there would be delays to service launch and excessive costs arising from:

- launching products and services without performing end-to-end PSTN/ISDN Platform integration and VV&T,
- separate teams duplicating the generic PSTN/ISDN Platform integration processes on several reference environments,
- no follow-through once designs have been partitioned into functional domains,
- no end-to-end process validation.

By comparison, platform integration offers:

- risk management across a number of products and services on each release of the PSTN/ISDN Platform (reduced time to market),
- economy of scale in supporting VV&T and integration of several PSTN/ISDN services with a single team on a single test environment (cost reduction),
- tracking of design and development activities through functional domains to ensure continued compliance with end-to-end product requirements (ensures fitness for purpose),
- end-to-end process validation (to prevent operational problems during or after product launch).

It is envisaged that this approach will eventually resemble a PSTN/ISDN production line, with several products being integrated for each release of the PSTN/ISDN platform. One important feature of this approach is that it ensures that PSTN/ISDN product teams benefit from the dedicated support of a product integration manager throughout the product life cycle, who has available the full range of expertise and resources of an experienced PSTN/ISDN Platform integration team.

The enterprise configuration management set of processes and tools [4] will provide a very important basis for this approach, as will the planned extension of the OMC-switch and CSS-OMC test environments. The Number Portability integration team has already piloted the CSS Unified Test Environment and this is also likely to be an important factor in future work.

9. Conclusions

This paper has described BT's current approach to number portability platform integration and test. It has also explained how this approach is now being reused on other PSTN/ISDN Platform products. It has emphasised the importance of testing operational processes with the systems which support them, and has described the benefits of applying the approach developed for number portability to expedite delivery while managing risk for similar PSTN/ISDN products and services.

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Steve Marsden joined BT in 1973, working on exchange maintenance and transmission system installation until he graduated with a BSc (Hons) in Electronics in 1984. Hethen became involved in a wide range of systems integration projects, and chaired the Network Management Forum Testing Team for several years, before becoming VV&T Manager on the Cyclone project.

He now leads a team responsible for Number Portability Systems Integration, Card Services Platform order entry VV&T, the integration of several products on the PSTN/ISDN platform, and messaging and groupware integration.